

TermPlanMT: Graph-Grounded Terminology Planning for French–English Regulatory Medical Document Translation

When Ontology Grounding Helps and When It Does Not

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Abstract. Terminology consistency is a critical quality dimension in the machine translation of regulatory medical documents, where mistranslated adverse event terms carry direct patient-safety consequences. We present **TermPlanMT**, a graph-augmented terminology-planning framework evaluated on a 126-segment French–English corpus drawn from a European Medicines Agency (EMA) Summary of Product Characteristics (SmPC). Six systems form an ablation ladder isolating the contributions of NER recall, MedDRA ontology grounding, and constrained decoding. Our central finding is that Concept Coverage Rate (CCR), the fraction of source spans grounded in a MedDRA ontology graph, is the binding constraint on terminology-aware MT: at 39% CCR, graph-free systems (S1–S2) outperform all graph-augmented systems on fluency – S1 leads on BLEU, S2 on chrF and Hypothesis–Reference Agreement (HRA) – while improving NER quality yields ≈ 7 BLEU gains for graph-augmented Mistral systems (S3–S5-Mistral) but not for NLLB or S2. HTM scores the certified human reference at 0.15–0.18, unchanged by full MedDRA alias expansion, confirming a structural mismatch between MedDRA nomenclature and professional translation practice. Ambiguous grounding causes the planning stage to abstain from locking critical terms. Dual human annotation on nine audit segments with LLM-assisted tie-breaking raises best-system agreement from 44% (raw) to 67% (adjudicated).

1 Introduction

EMA Summaries of Product Characteristics (SmPCs) require a single adverse-event concept to receive one identical English rendering across an entire document—a requirement that affects pharmacovigilance signal detection and patient safety [?]. Neural MT handles medical prose fluently but is structurally indifferent to cross-segment consistency, producing term drift (same source term, multiple renderings), concept flattening (Lowest Level Term (LLT)-level terms collapsed to hypernyms), and coverage gaps (multi-token spans outside NER). MedDRA’s five-level hierarchy (Table 1) provides validated English labels that are an attractive grounding resource—if the grounding pipeline can cover enough of the source text. This paper makes the following contributions:

1. **A six-system ablation ladder** that isolates the individual contributions of NER recall, MedDRA ontology grounding via Neo4j, and constrained decoding enforcement to terminology-aware MT quality.
2. **Three novel evaluation metrics** – HTM, rHTM, and HRA – that measure terminology alignment at the ontology level rather than surface string overlap, with a diagnostic showing $rHTM \approx 0.15\text{--}0.18$ for certified human references.
3. **A CCR bottleneck analysis** demonstrating that NER extraction quality, not retrieval or decoding strategy, is the primary constraint on graph-augmented MT performance.
4. **A mechanistic consistency analysis** explaining the paradox in which graph-augmented systems produce lower cross-document term consistency than graph-free baselines, traced to the planning stage’s abstain-on-ambiguity policy.
5. **Inter-annotator reliability study** on nine audit segments with dual human labels, LLM-assisted tie-breaking, and adjudicated best-system agreement rising from 44% to 67%.
6. **Two named phenomena:** the **register mismatch** (graph-augmented systems produce

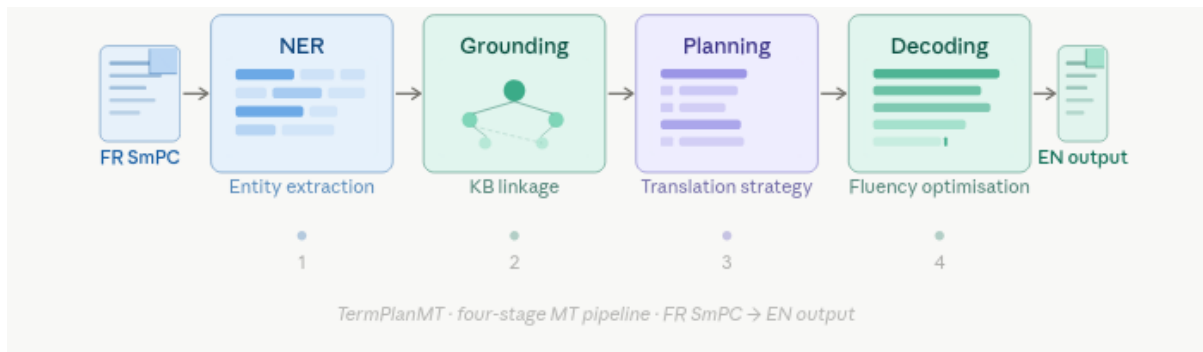


Figure 1: Four-stage TermPlanMT pipeline (NER → grounding → planning → decoding).

pharmacovigilance coding language while certified translators write clinical prose) and the **consistency paradox** (the most ontology-correct system is the least human-like and is never the preferred output in 30-segment human evaluation).

2 Related Work

Lexically constrained decoding [??] and terminology-aware training [?] enforce surface-level term appearance but ignore ontological hierarchy and do not address candidate selection when multiple levels are valid. ? show that feeding long context directly into NMT degrades output quality due to attention mechanism limitations, motivating our pre-translation planning stage: rather than exposing the full document at decoding time, we distil relevant terminology into a compact, structured prompt. ? release BioMistral, the medically adapted LLM used for NER here. ? identify biomedical named entities broadly—diseases, drugs, and procedures—as a persistent failure mode in biomedical MT. Our work narrows this to the specific subtype of adverse event (AE) terminology as defined by the MedDRA ontology, a distinction with direct regulatory consequences given that AE terminology is legally standardised across EMA submissions. GraphRAG-style retrieval [?] into LLM prompts provides the template for our S3 system, adapting knowledge-grounded MT [?] to a Neo4j MedDRA backend.

3 Dataset

Corpus.. Section 4.8 (Undesirable Effects) of the KEYTRUDA (pembrolizumab) SmPC [?]: 126 aligned French–English segment pairs after excluding segment 48_028 (a dense frequency table whose n-gram overlap distorts corpus metrics). Coverage: $\approx 4,400$ words; all three failure modes at high density.

MedDRA (Medical Dictionary for Regulatory Activities) represents adverse events using a five-level hierarchical terminology structure, enabling both fine-grained clinical coding and higher-level safety aggregation. In our corpus, adverse-event mentions map almost exclusively to L5 ($\approx 96\%$).

Two NER conditions (zero-shot vs. QUAERO fine-tuned).. We hold the MT stack (S1–S6) fixed and vary only how French spans are extracted before Neo4j grounding. Both conditions use **BioMistral-7B** [?]; they differ in prompting versus supervised adaptation on biomedical French NER.

Condition A — zero-shot prompting (ner_biollm): BioMistral-7B in 4-bit inference with a French JSON-list prompt (no weight updates). On the 126-segment evaluation set this yields CCR = 39.0% (184 grounded spans out of 472 extracted). The run also shows a systematic prompting artefact: *hypothyroïdie* appears in 92/127 full-corpus segments regardless of clinical content, which partially contaminates cross-segment consistency scores for systems that consume these spans.

Condition B — Unsloth LoRA on QUAERO (ner_biollm_ft): the same base model fine-tuned with **Unsloth** on the **QUAERO** French Medical corpus in BRAT format (EMEA + MEDLINE train splits; **1,540** sentence-level training examples; entity types DISO, CHEM, PROC only). Each sentence is an Alpaca instruction example (extract disorders, drugs, and procedures as a JSON string list); training uses 4-bit loading, LoRA rank 16 / $\alpha=32$ / dropout 0.05 on all linear projections, 3 epochs, per-device batch 4, gradient accumulation 4, learning rate 2×10^{-4} , cosine schedule, 10 warmup steps, and response-only loss (SFTTrainer). Spans are written to

Table 1: MedDRA hierarchy levels (MedDRA v28) with acronym, meaning, and specificity; examples follow the Section 4.8 immune-mediated pneumonitis chain.

Level	Abbr.	Meaning	Specificity	Example
L1	SOC	System Organ Class	Broadest	Respiratory, thoracic and mediastinal disorders
L2	HLGT	High Level Group Term	Broad	Lower respiratory tract inflammatory and immunologic conditions
L3	HLT	High Level Term	Intermediate	Pneumonitis and lung infiltration disorders
L4	PT	Preferred Term	Specific clinical concept	Pneumonitis
L5	LLT	Lowest Level Term	Most specific	Immune-mediated pneumonitis

segments_ner_unsloth_full.jsonl and fed through the identical pipeline. On the evaluation set: 211 grounded spans out of 580 extracted (CCR = 36.4%)¹; boundaries are cleaner and the hypothyroidism hallucination disappears.

Graph-augmented Mistral systems (S3–S5–Mistral) gain ≈ 7 BLEU only under Condition B (Table 5), while S1/S2 translations are unchanged. That pattern isolates *span quality*—usable grounding and clean boundaries—as the binding constraint upstream of graph retrieval, not the graph machinery itself.

4 System Architecture

Six systems form an ablation ladder (Table 2). S1–S5 share the same four-stage pipeline: French NER → MedDRA grounding via Neo4j → global terminology lock → MT decoding. S6 is a glossary oracle ablation that replaces the MedDRA graph with a curated FR→EN glossary extracted from the reference translations.

ID	Model	Graph	Enforce.
S1	NLLB-200 600M	No	None
S2	Mistral-7B (doc)	No	None
S3	Mistral+GraphRAG	Yes	None
S4	S3 + reranking	Yes	Soft
S5	NLLB/Mistral+logit	Yes	Hard
S6	NLLB + glossary	Oracle	Hard

Table 2: Ablation ladder. S1 = NMT lower bound; S2 = LLM with full document context; S3 is the key novel condition; S6 isolates term-source quality from graph retrieval.

¹CCR depends on both NER recall and whether a span exists in the MedDRA graph. Condition B extracts more spans (580 vs. 472) but grounds a larger share of difficult/rare strings, so the *rate* can fall while the absolute grounded count rises (211 vs. 184). Downstream comparisons should treat CCR as a diagnostic, not as the manipulated variable.

Terminology planning.. Before translation, `compute_global_locks()` scores each candidate English rendering c :

$$\text{Score}(c) = \alpha \text{Ctx}(c) + \beta \text{Nbr}(c) - \gamma \text{HP}(c) \quad (1)$$

where Ctx is cosine similarity to source segments containing the term, Nbr is similarity to adjacent MedDRA concept renderings, and $\text{HP} = |\hat{d} - d_{\text{src}}|$ penalises wrong specificity level ($\alpha=0.5$, $\beta=0.3$, $\gamma=0.2$). When a term matches *multiple* MedDRA concepts, no lock is committed (§8). For each unambiguous grounded term, the planner scores candidate English renderings from the concept node and its graph neighbours, commits the highest-scoring label, and skips ambiguous keys entirely.

The abstain-on-ambiguity policy is a deliberate safety choice: when a French term maps to multiple MedDRA concepts with equal confidence (e.g. *pneumopathie inflammatoire* maps to both PNEUMONITIS and INTERSTITIAL LUNG DISEASE), forcing a lock risks propagating the wrong concept across the entire document. The cost of this conservatism is reduced cross-document consistency, since unlocked terms are resolved independently at each segment by the decoder. We discuss this consistency paradox in Section 8.

S5 – Logit Boosting with Hard Fallback.. S5 applies a two-layer decoding strategy. First, a soft logit bias (PhraseLogitBoost, $\delta=1.75$ for NLLB, $\delta=1.25$ for Mistral) increases the probability mass of tokens belonging to locked MedDRA renderings without preventing the model from generating alternative phrasings if contextually motivated. Second, for NLLB only, `force_words_ids` enforces the locked phrase as a hard constraint via constrained beam search, with automatic fallback to soft-only decoding if the constraint causes beam failure. The Mistral variant uses soft boosting only, since instruction-tuned

LLMs are more sensitive to hard token forcing and exhibit greater fluency degradation under strict constraints.

4.1 MedDRA Knowledge Graph Construction

The grounding component $\mathcal{G}$ is a Neo4j property graph built from MedDRA v28 flat files (academic licence via UPV/EHU): `mdhier.asc` rows yield $\approx 80,000$ LLT records with five-level lineage; French `fr_label` properties are merged with case folding. **7,679** normalised French keys are ambiguous (multiple concepts); these trigger planner abstention. The loaded graph has **84,404 nodes** and **91,082 PT/LLT fuzzy-index entries**. Runtime grounding uses exact match then RapidFuzz ($\geq 90\%$) on PT/LLT nodes only; `neighbours()` and `same_branch()` support planning scores. CCR diagnostics on baseline NER: 23% exact, 11% fuzzy, 6% ambiguous, 60% not-in-graph (mostly drugs, grades, and numerics outside the AE subset).

5 Metrics

Metric definitions.

- **HTM** (Hierarchy-Aware Terminology Metric): per grounded NER span, score = 1.0 (exact MedDRA level), 0.5 (same branch), 0.0 (miss); macro-averaged over grounded spans.
- **rHTM** (Reference HTM): HTM applied to the certified human reference, not a system hypothesis. A dataset-level ceiling; values of 0.15–0.18 confirm translators prefer phrasings outside the MedDRA preferred-term vocabulary.
- **HRA** (Hypothesis–Reference Agreement): $1 - |\text{HTM}_{\text{hyp}} - \text{HTM}_{\text{ref}}|$, macro-averaged. Measures agreement with professional translation practice rather than raw ontology proximity.
- **HTM-A** (Alias-Expanded HTM): accepts any ancestor up to SOC. Score invariance across tiers T1–T3 confirms failures are conceptual, not surface-level.
- **CCR** (Concept Coverage Rate): grounded spans / extracted spans. Identical across S1–S6 for a given NER condition; the primary diagnostic for NER quality.

Fluency: corpus BLEU [?] and chrF++ [?] via SacreBLEU; neural: COMET [?]. HTM-A extends HTM matching across three tiers: T1 exact LLT; T2 adds Preferred Term (PT) parent; T3 adds all same-branch ancestors.

CCR and drift.. CCR = grounded/extracted spans; identical across S1–S6 for a given NER condition. Term drift = $(D - 1)/(N - 1)$ for D

distinct renderings across N segments; drift = 0 is perfect consistency.

6 Results

6.1 Main Results

System	BLEU	chrF++	COMET	HTM	HRA
S1 NLLB	22.89	37.14	0.639	0.321	0.672
S2 Mistral	22.51	39.95	0.637	0.413	0.743
S3 GraphRAG	10.39	34.36	0.586	0.397	0.601
S4 rerank	9.93	34.24	0.583	0.429	0.579
S5 NLLB+boost	21.59	35.91	0.638	0.353	0.661
S5 Mist+boost	10.32	34.32	0.585	0.380	0.607
S6 NLLB+gls*	20.00	35.11	0.634	0.321	0.694
Reference	–	–	–	0.151	1.000

Table 3: Baseline NER condition (`ner_biollm`, CCR = 39.0%; 126 segments, 48_028 excluded). BLEU/chrF++ via SacreBLEU; COMET via wmt22-comet-da; HTM/HRA from Neo4j. *S6 is a glossary oracle (terms from reference).

System	BLEU	chrF++	COMET	HTM	HRA
S1 NLLB	22.89	37.14	0.639	0.550	0.429
S2 Mistral	22.51	39.95	0.637	0.735	0.476
S3 GraphRAG [†]	19.58	35.33	0.626	0.664	0.440
S4 rerank [†]	19.66	35.36	0.626	0.664	0.450
S5 NLLB+boost	21.37	36.06	0.637	0.578	0.403
S5 Mist+boost [†]	20.61	35.78	0.626	0.673	0.450
S6 NLLB+gls*	20.09	34.81	0.632	0.536	0.408
Reference	–	–	–	0.178	1.000

Table 4: Fine-tuned NER condition (`ner_biollm_ft`; CCR = 36.4%; 126 segments, 48_028 excluded). [†]Contamination-filtered (18/127 rows scored as empty hyps). *S6 uses a glossary oracle from the reference.

Tables 3 and 4 report corpus-level scores; Fig. 2 visualises the same systems (baseline BioMistral NER left, fine-tuned NER right). Under fine-tuned NER, graph-augmented Mistral systems close the fluency gap substantially, confirming NER quality as the binding constraint.

HTM is recall-dependent on NER quality..

BLEU and chrF scores for S1 and S2 are identical across both NER conditions (Tables 3 and 4), confirming that the underlying translations did not change. However, HTM scores shift substantially: S1 HTM rises from 0.321 to 0.550 and S2 from 0.413 to 0.735 under fine-tuned NER. This is a *metric artifact*, not a translation improvement. HTM is

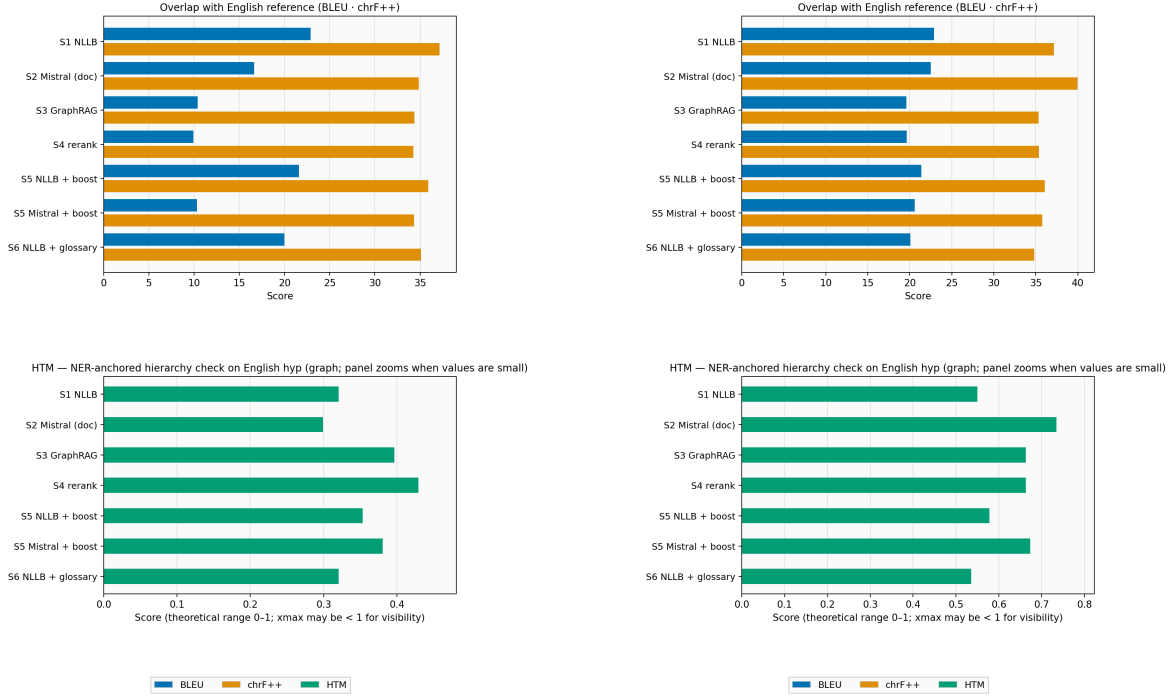


Figure 2: Grouped metrics by system. **Left:** baseline NER (ner_biollm, CCR = 39.0%). **Right:** fine-tuned NER (ner_biollm_ft, CCR = 36.4%). Top: BLEU and chrF++; bottom: HTM (lexical, graph-aware). HTM is not comparable across panels.

computed only over spans the NER model successfully extracts; a stronger NER model exposes a larger and more challenging set of terminology spans, changing both the denominator and the matched set simultaneously. Scores across NER conditions are therefore not directly comparable and should be interpreted within each condition only.

6.2 CCR is the Binding Constraint

Table 5 isolates the effect of NER quality by holding all other pipeline components fixed and comparing the same system across NER conditions.

S1 is unchanged (no graph). S2 is exactly zero: the same inference JSONL is shared across both NER conditions, so ΔBLEU is zero by construction. S3–S5-Mistral show ≈ 7 -point ΔBLEU with intervals strictly above zero, isolating the NER bottleneck on the graph path. S5-NLLB is effectively flat. The S6 oracle glossary yields negligible gain, showing that logit boosting with reference-extracted terms does not improve over the graph path once NER quality is matched.

The 95% paired bootstrap confidence intervals for the BLEU gain of fine-tuned graph-augmented Mistral systems (S3–S5-Mistral) over

System	ΔBLEU	$\Delta\text{chrF++}$	95% CI ΔBLEU
S1 NLLB	± 0.00	± 0.00	[0.00, 0.00]
S2 Mistral [‡]	± 0.00	± 0.00	[0.00, 0.00]
S3 GraphRAG	+7.07	+0.97	[+1.07, +11.10]
S4 rerank	+7.42	+1.12	[+1.18, +11.53]
S5 NLLB+boost	−0.16	+0.15	[−0.98, +0.66]
S5 Mist+boost	+7.16	+1.46	[+1.10, +11.66]
S6 NLLB+gls [*]	+0.09	−0.30	–

Table 5: Paired gain (fine-tuned minus baseline NER) on 126 segments. ΔBLEU is the paired bootstrap estimate (2,000 resamples, seed 42); 95% CIs for S3–S5-Mistral exclude 0. $\Delta\text{chrF++}$ is the corpus-level paired difference. [‡]S2 shares the same inference JSONL across conditions. ^{*}S6 CI not computed (oracle glossary).

their baseline-NER counterparts exclude zero (see Table 5), confirming statistical significance at $p < 0.05$. Graph-free systems (S1, S2) and S5-NLLB show confidence intervals that include zero, indicating their apparent gains are not reliably distinguished from noise.

S1 leads on corpus BLEU and S2 leads on chrF and HRA in the baseline NER condition (Table 3): graph-augmented systems fall well below both because the graph covers fewer than 40% of source

spans while injected MedDRA strings hurt fluency in segments where it is active. S2’s lead on HRA (0.743) reflects the human-like output of full-document Mistral context without graph interference. The contrast between zero Δ BLEU for S1, S2, and S5-NLLB versus large Δ BLEU for S3–S5-Mistral isolates the bottleneck upstream of graph retrieval. Higher NER coverage would likely close the S2/S3 fluency gap.

A post-hoc CCR diagnostic (tools/ccr_diagnostic.py) classifying the 171 unique baseline NER terms against the Neo4j graph found that 60% are absent from MedDRA entirely (NOT_IN_GRAPH), compared to 54% of the 246 fine-tuned terms; a further 6–8% are ambiguous (multiple concept matches, triggering the planner’s abstain policy), and only 23–32% ground via exact string match. The dominant not-in-graph category is populated by non-medical tokens (*pembrolizumab*, grade annotations, numeric ranges) that NER extracted as candidate spans but that have no MedDRA equivalent – confirming that the CCR ceiling is set by NER precision, not by graph coverage gaps.

6.3 Missing Terms and Consistency

Under baseline NER, S5-Mistral achieves the lowest missing-term rate; S1-NLLB misses the majority of grounded terms. Cross-document modal agreement is highest for graph-free systems (S1: 0.993, S2: 0.984); graph-augmented systems recover under fine-tuned NER but do not overtake S2 (0.985).

6.4 Inter-Annotator and Audit Agreement

Nine Section 4.8 segments were double-annotated by two human raters (D, N) on register (binary), pattern acceptability (S1, S2, S5, S6), and best_system choice. An LLM-assisted audit supplied reference labels for *tie-breaking only*, not as an equal third judge. Where raters tied between systems in free-text notes, adjudication followed documented rules: prefer the LLM label when it lies in the tied set; otherwise prefer N; if the LLM is unknown, use N (supplementary adjudication_decision_table.csv).

Annotator N aligned more closely with the LLM audit on pooled pattern labels ($\kappa=0.52$) than D ($\kappa=0.36$), while adjudicated D matched the LLM on best-system in 7/9 cases. Pattern fields showed the largest human–human splits (e.g. S2/S6 $\kappa < 0$ for D vs N), consistent with register and paraphrase disagreements documented in per-segment notes.

Comparison	Agree	%	κ
<i>Best-system choice (9 segments)</i>			
D vs N (raw)	4/9	44	0.00
D vs N (adjudicated)	6/9	67	0.18
D vs LLM (adjudicated D)	7/9	78	0.60
N vs LLM	4/9	44	0.00
<i>All fields pooled (54 binary labels)</i>			
D vs N	–	44	0.15
D vs LLM	–	59	0.36
N vs LLM	–	69	0.52

Table 6: Human–human and human–audit agreement. Raw best-system disagreement reflects genuine preference splits (N favours S2 in 7/9 segments); adjudication raised D–N best-system agreement from 44% to 67%. $\kappa=0$ at 44% best-system agreement means observed matches equal chance agreement given skewed marginals ($n=9$).

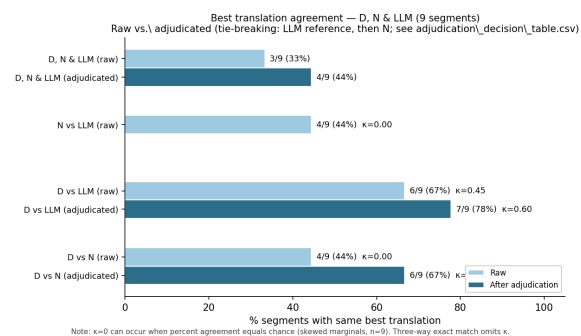


Figure 3: Best-system agreement: raw vs. adjudicated bars. Segments 1, 5, 6, and 9 required adjudication; 2, 3, 4, 7, and 8 had consensus or single-reference resolution.

7 The HTM–Practice Gap

7.1 Why the Reference Scores Near Zero

On the baseline NER manifest, $r_{\text{HTM}} = 0.151$ on the professional reference (Table 3); under fine-tuned NER the reference HTM is 0.178 (Table 4) because the grounded span set differs. In both cases the score stays below several MT systems on HTM. Professional translators do not transcribe MedDRA LLT strings verbatim: the LLT *Immune-mediated pneumonitis* appears as *pneumonitis* in tabular contexts and as *immune-related pneumonitis* in descriptive contexts.

Tier	Accepts	Low	Hi
T1	Exact LLT	0.151	0.178
T2	T1 + PT parent	0.151	0.178
T3	T2 + branch ancestors	0.151	0.178

Table 7: rHTM-A on the professional reference under three alias tiers (Low = baseline NER manifest; Hi = fine-tuned NER manifest). All tiers yield identical scores within each manifest.

7.2 Alias Expansion Does Not Close the Gap

The ancestor chain for *pneumopathie inflammatoire* contains *pneumonitis*, so the null T1→T3 delta is not a graph coverage gap. Professional translators use phrasings such as *immune-related pneumonitis* and *inflammatory lung disease* that lie entirely outside the MedDRA preferred-label tree at any level. The mismatch is structural: MedDRA labels are designed for pharmacovigilance coding, not regulatory prose.

HTM should therefore be interpreted as a measure of MedDRA-proximity, not translation quality. The HRA column is the complementary measure. Notably, graph-augmented systems score consistently *lower* on HRA than S2 (Table 3: S3 = 0.601 vs. S2 = 0.743; Table 4: S3 = 0.440 vs. S2 = 0.476): the graph pushes outputs toward MedDRA labels that professional translators avoid. Per-system inference times are in [results/ner_biollm/figures/](#).

8 Consistency Failure Analysis

Term drift (qualitative).. *Pneumopathie inflammatoire* occurs in four baseline segments where it is grounded for S3/S4/S5-Mistral, and in eight fine-tuned segments (more NER hits for the same French string). Those systems emit two renderings—often *pneumonitis* in incidence-style sentences and *immune-mediated pneumonitis* elsewhere—yielding drift scores of 1/3 (baseline) and 1/7 (fine-tuned). S1 collapses to one rendering for this span in both conditions (drift 0). S2 is stable in baseline (drift 0) but shows non-zero drift under fine-tuned NER once the span appears in additional contexts. S3/S4/S5-Mistral thus exhibit the clearest cross-segment inconsistency for this exemplar.

Mean modal agreement (§6) shows the same pattern at corpus level: graph-free systems rank highest under baseline NER; graph-augmented systems recover partially under fine-tuned NER but

do not overtake S2.

Translator language outside MedDRA.. Professional translators systematically prefer descriptive phrasings outside the MedDRA preferred-term vocabulary, falling into three patterns: **descriptive expansion** (e.g. “*immune-related pneumonitis*” for the PT PNEUMONITIS, scoring 0.0 under HTM at any alias tier); **hypernym substitution** (deliberate ascent to HLT/SOC for readability, penalised as a specificity error by HTM); and **compound paraphrase** (clinical shorthand such as “*hand-foot syndrome*” lying outside the MedDRA branch entirely). Systems scoring well on HTM therefore use formal ontology labels that certified translators avoid, explaining the HTM-BLEU divergence. The appropriate evaluation lens is **HRA**, which measures agreement with professional translation practice rather than raw ontology proximity.

Unified expert evaluation.. A GPT-4o EMA-translator persona ranked eight outputs on 30 segments ([error_analysis/unified_eval.csv](#)). The human reference won 20/30 best-output votes (67%); S2 ranked best among MT systems (mean rank 3.27); S5 received zero best-output votes. S1’s *daxitinib* for *axitinib* substitution was flagged as a critical pharmacovigilance error.

Mechanism.. *Pneumopathie inflammatoire* maps to two distinct MedDRA concepts (one of 7,679 ambiguous normalised keys in the graph). The planner commits no lock for ambiguous terms, so S3/S4/S5-Mistral resolve the term context-locally per segment, producing drift scores of 1/3 (baseline) and 1/7 (fine-tuned). S1-NLLB produces zero drift not through terminological reasoning but through vocabulary rigidity. The fix is *soft ambiguity resolution*: commit the top-scoring candidate as a provisional lock for human review, consistent with CAT-tool workflows.

Output contamination.. In the fine-tuned NER condition, 18/127 segments (14.2%) contain prompt contamination from graph-context Mistral runs; filtered scores are used throughout (S3 BLEU 15.36→19.58 after filtering).

9 Discussion and Limitations

The central finding is a paradox. S5 is more technically correct than the human reference by ontology standards – scoring higher on HTM –

yet received zero best-output votes across 30 segments in the unified expert evaluation. The human reference dominated at 67%; **S2 (Mistral, full document context, no graph) is the best MT system** with an average expert rank of 3.27, ahead of S3-FT (4.69). S3-FT was competitive in an earlier pilot on a different 25-segment sample but does not replicate that advantage on the canonical sample. MedDRA labels are designed for pharmacovigilance database coding, not clinical prose; a translator writing “*immune-related pneumonitis*” is writing for a reader, while S5 writing PNEUMONITIS is writing for a database. Both are technically correct but serve different purposes, and optimising for HTM targets the wrong register.

The graph works, but only upstream of a CCR threshold. At 39% CCR the graph hurts fluency more than it helps terminology. Under fine-tuned NER (211 grounded spans vs. 184 baseline), graph-augmented Mistral systems gain ≈ 7 BLEU with bootstrap intervals excluding zero. The bottleneck was never graph retrieval or decoding strategy – it was always NER coverage. The S6 oracle glossary confirms this: even with reference-extracted terms, BLEU improves by only +0.09 points over S5-NLLB.

Limitations.. *Single document and reference:* all results derive from one SmPC section against one professional translation. *HTM ceiling:* rHTM is low (0.15–0.18) and invariant to alias expansion, so HTM is only interpretable relatively. *NER artefact:* baseline NER contains hallucinated spans (*hypothyroïdie* in 92/127 segments). *Contamination:* 14.2% of fine-tuned Mistral graph outputs contain prompt contamination; filtered scores are used throughout. *Boost surface rate:* the lock map was not logged during inference; proxy analysis shows S5 surfaces NER terms verbatim in 34% of cases overall, dropping to 0% for multi-token MedDRA labels, confirming that term length is the hard ceiling on any enforcement approach.

10 Conclusion

TermPlanMT demonstrates that CCR is the binding constraint on graph-augmented regulatory MT: at 39% CCR, graph-free systems outperform all graph-augmented systems on fluency – S1 leading on BLEU, S2 on chrF and HRA – while improving NER quality yields ≈ 7 -point corpus BLEU gains for Mistral graph systems with bootstrap intervals excluding zero. A unified expert evalua-

tion (30 segments, 8 systems, GPT-4o EMA translator persona) confirms that **S2 is the best MT system** by average expert rank (3.27), with the human reference dominating at 67% best-output votes. S5 received zero best-output votes in both NER conditions: the most ontology-correct system is never the preferred output, because MedDRA labels serve pharmacovigilance coding rather than clinical prose. Three broader findings emerge: MedDRA nomenclature and professional translation practice are structurally misaligned in ways alias expansion cannot fix; ambiguous grounding produces predictable term drift through the planning stage’s abstain-on-ambiguity policy; and NER quality is a patient-safety concern – S1’s hallucination of *daxitinib* for *axitinib* illustrates the risk of deploying low-quality NER in pharmacovigilance contexts. Substantially higher CCR with cleaner NER span boundaries is the necessary condition for graph-augmented systems to surpass S2; quantifying that threshold is left to future work with broader French biomedical NER.

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